

Code: 237

PART 1: MULTIPLE CHOICE QUESTIONS (30 points)

Answer all questions by choosing **ONE** correct option (A, B, C, or D).

Q1

The rental price of capital is:

- A. the price paid to use capital for a limited time period
 - B. always more than the purchase price
 - C. determined outside the realm of factor markets
 - D. the price paid for ownership of the capital
-

Q2

Writing total output as Q , change in output as ΔQ , employment as L , and change in labor as ΔL , the marginal product of labor is:

- A. $\Delta L/\Delta Q$
 - B. Q/L
 - C. $\Delta Q/\Delta L$
 - D. ΔQL
-

Q3

A monopoly is an inefficient way to produce a product because:

- A. it produces a smaller output than in a competitive market
 - B. it faces a downward-sloping demand curve
 - C. it can earn both short-run and long-run profits
 - D. the cost of producing one more unit exceeds its value to buyers
-

Q4

Average total cost equals:

- A. average variable cost + total fixed cost
 - B. total cost – total quantity of output
 - C. total cost / output
 - D. output / total cost
-

Q5

Which formula correctly calculates a monopolist's profit?

- A. profit = price – marginal cost
 - B. profit = (price – marginal cost) × quantity
 - C. profit = price – average total cost
 - D. profit = (price – average total cost) × quantity
-

Q6

An optimal tax on pollution would be one in which:

- A. producers internalize the cost of pollution
 - B. a planner maximizes production
 - C. consumer value exceeds production cost including tax
 - D. producers create no pollution
-

Q7

A function showing maximum output from given inputs is called:

- A. production possibility curve
 - B. isoquant
 - C. production function
 - D. isocost function
-

Q8

Which statement is NOT correct?

- A. Variable costs change as output changes
 - B. Fixed costs are constant
 - C. Average fixed costs are constant
 - D. Average total costs are typically U-shaped
-

Q9

In perfect competition, a firm maximizes profit by choosing:

- A. output where price equals marginal cost
 - B. output where AVC is minimized
 - C. output where AFC is minimized
 - D. price where MC equals ATC
-

Q10

If wheat price decreases, what happens to flour supply?

- A. supply of flour increases
 - B. demand for flour decreases
 - C. supply of flour decreases
 - D. demand for flour increases
-

Q11

Which is an implicit cost?

- A. foregone rent on owned office space
 - B. cash paid for raw materials
 - C. salaries to owners
 - D. interest on loans
-

Q12

Oligopolists earn highest profit if they:

- A. charge monopoly price
 - B. follow self-interest
 - C. produce competitive quantity
 - D. produce more than competitive quantity
-

Q13

In perfect competition, price equals:

- A. total revenue
 - B. efficient scale
 - C. average revenue
 - D. marginal revenue
-

Q14

The monopolist's supply curve:

- A. is vertical
 - B. does not exist
 - C. is horizontal
 - D. is upward sloping
-

Q15

Monopolistic competition firms charge higher prices because:

- A. decreasing returns to scale
 - B. increasing costs
 - C. $MR = MC$
 - D. downward-sloping demand
-

Q16

Price elasticity of demand is:

- A. $\% \Delta Q / \% \Delta P$
- B. $\% \Delta \text{Income} / \% \Delta Q$

- C. $\% \Delta P / \% \Delta Q$
 - D. $\% \Delta Q / \% \Delta \text{Income}$
-

Q17

A firm shuts down short run when:

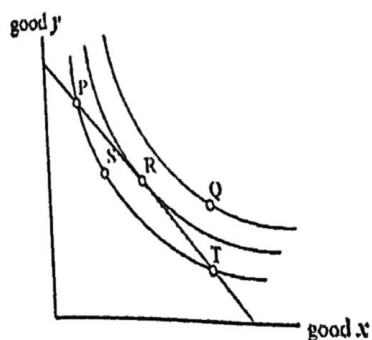
- A. $P < ATC$
 - B. $AR > AFC$
 - C. $P < AVC$
 - D. $AR > MC$
-

Q18

Law of demand says:

- A. lower price increases demand
 - B. lower price increases quantity demanded
 - C. higher price increases quantity demanded
 - D. higher price decreases demand
-

Q19



Optimal affordable bundle is point:

- A. S
 - B. Q
 - C. P
 - D. R
-

Q20

Impact of one person on another is called:

- A. economic dilemma
 - B. externality
 - C. deadweight loss
 - D. third-party problem
-

Q21

If goods are substitutes, cross-price elasticity is:

- A. one
 - B. negative
 - C. zero
 - D. positive
-

Q22

A firm hires labor until value of marginal product equals:

- A. AVC
 - B. price
 - C. wage
 - D. ATC
-

Q23

Duopoly statements:

- A. collusion equals monopoly output
 - B. self-interest lowers price below monopoly
 - C. self-interest raises output above monopoly
 - D. all are correct
-

Q24

At profit-maximizing output:

- A. $MR = MC$
 - B. $MR = ATC$
 - C. $AR = ATC$
 - D. $MR = AVC$
-

Q25

A dominant strategy is:

- A. best regardless of others
 - B. best if others cooperate
 - C. dominates others
 - D. in every game
-

Q26

Most likely monopoly:

- A. online bookstore
 - B. local electric cooperative
 - C. local restaurant
 - D. national florist
-

Q27

Highest income elasticity:

- A. water
 - B. diamonds
 - C. hamburgers
 - D. housing
-

Q28

Higher concentration ratio means:

- A. more firm control
 - B. more competitive
 - C. less competitive
 - D. A and C
-

Q29

Not competitive market feature:

- A. identical products
 - B. limited entry
 - C. price takers
 - D. profit maximization
-

Q30

If monopolistic firms earn profit, long run:

- A. profits stay
- B. consumers leave
- C. new firms enter, profits fall
- D. government regulates

Part 2: Problem Solving (10 points)

Problem 1

Given the demand and supply functions for the product as follows:

$$Q_d = 200 - 3P$$

$$Q_s = 10P - 20$$

- a. Determine the market equilibrium price and quantity. Illustrate by a demand-supply diagram.
- b. Calculate the price elasticity of demand at the market equilibrium price. State the meaning of the result.

- c. If the government sets the product's price at \$6, will the market experience a **surplus** or a **shortage**? By how many units?
- d. The government imposes a tax of \$1 per unit on sellers. Determine the new equilibrium price and quantity.
-

Problem 2

The demand and total cost functions for a company are given as follows:

$$P = 190 - 0.5Q$$

$$TC = 4000 + 10Q + 0.5Q^2$$

- a. Write the following cost functions: **ATC**, **VC**, **AVC**, **MC**.
- b. What price should the company charge if it wants to maximize its profit in the short run? Calculate the company's total profit.
- c. What price should it charge if it wants to maximize its revenue in the short run? Calculate the company's total revenue.
- d. Find the firm's break-even output and price (where profit equals zero).
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Code: 348

Part 1: Multiple Choice Questions (30 points)

Choose the correct answer (A, B, C, or D) for each question.

Q1

In a game, a dominant strategy is:

- A. the best strategy for a player to follow, regardless of the strategies followed by other players
- B. the best strategy for a player to follow only if other players are cooperative

- C. a strategy that leads to one player's interests dominating the interests of other players
 - D. a strategy that must appear in every game
-

Q2

If two goods are substitutes, their cross-price elasticity of demand is:

- A. one
 - B. negative
 - C. zero
 - D. positive
-

Q3

The impact of one person's actions on the well-being of a bystander is called:

- A. an economic dilemma
 - B. an externality
 - C. deadweight loss
 - D. the third-party problem
-

Q4

At the profit-maximizing level of output:

- A. marginal revenue equals marginal cost
 - B. marginal revenue equals average total cost
 - C. average revenue equals average total cost
 - D. marginal revenue equals average variable cost
-

Q5

For a firm in a perfectly competitive market, the price of the good is always:

- A. equal to marginal revenue
- B. equal to the firm's efficient scale of output
- C. equal to total revenue
- D. greater than average revenue

Q6

Which of the following would be most likely to have monopoly power?

- A. an online bookstore
 - B. a local electrical cooperative
 - C. a local restaurant
 - D. a national florist
-

Q7

A firm under monopolistic competition produces less and charges a higher price than under perfect competition because:

- A. the firm faces decreasing returns to scale
 - B. the firm operates where $MR = MC$
 - C. the firm faces increasing costs
 - D. the firm faces a downward-sloping demand curve
-

Q8

The law of demand states that, other things equal:

- A. when price falls, demand rises
 - B. when price falls, quantity demanded rises
 - C. when price rises, demand falls
 - D. when price rises, quantity demanded rises
-

Q9

Wheat is the main input in flour production. If wheat price decreases, we expect:

- A. supply of flour to increase
 - B. demand for flour to decrease
 - C. supply of flour to decrease
 - D. demand for flour to increase
-

Q10

Average total cost equals:

- A. total cost – total output
 - B. average variable cost + total fixed cost
 - C. total cost / output
 - D. output / total cost
-

Q11

Which formula correctly calculates a monopolist's profit?

- A. profit = (price – ATC) × quantity
 - B. profit = price – ATC
 - C. profit = price – MC
 - D. profit = (price – MC) × quantity
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Q12

A function showing maximum output from given inputs is called:

- A. isoquant
 - B. production possibility curve
 - C. production function
 - D. isocost function
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Q13

The monopolist's supply curve:

- A. is vertical
 - B. does not exist
 - C. is horizontal
 - D. is upward sloping
-

Q14

Which statement is correct about duopoly?

- A. Collusion gives monopoly output
 - B. Self-interest lowers price below monopoly
 - C. Self-interest raises output above monopoly
 - D. All are correct
-

Q15

Oligopolists earn highest profit if they:

- A. produce competitive quantity
 - B. produce more than competitive quantity
 - C. follow self-interest
 - D. charge monopoly price
-

Q16

A firm shuts down in the short run when:

- A. $P < ATC$
 - B. $AR > AFC$
 - C. $P < AVC$
 - D. $AR > MC$
-

Q17

The rental price of capital is:

- A. always more than purchase price
 - B. price paid to use capital for a limited time
 - C. outside factor markets
 - D. price paid for ownership
-

Q18

In a competitive market, profit is maximized when:

- A. $P = MC$
 - B. AFC is minimized
 - C. $MC = ATC$
 - D. AVC is minimized
-

Q19

Not a characteristic of perfect competition:

- A. identical products
 - B. price takers
 - C. limited entry
 - D. profit maximization
-

Q20

Marginal product of labor is:

- A. $\Delta L / \Delta Q$
 - B. Q / L
 - C. $\Delta Q / \Delta L$
 - D. $\Delta L / Q$
-

Q21

A firm hires labor until value of marginal product equals:

- A. AVC
 - B. price
 - C. wage
 - D. ATC
-

Q22

A monopoly is inefficient because:

- A. earns long-run profit
- B. produces less than competitive output

- C. $MC > \text{value to buyers}$
 - D. faces downward demand
-

Q23

Highest income elasticity:

- A. water
 - B. diamonds
 - C. hamburgers
 - D. housing
-

Q24

Price elasticity of demand is:

- A. $\% \Delta P / \% \Delta Q$
 - B. $\% \Delta Q / \% \Delta P$
 - C. $\% \Delta \text{Income} / \% \Delta Q$
 - D. $\% \Delta Q / \% \Delta \text{Income}$
-

Q25

Optimal pollution tax makes:

- A. planner maximize output
 - B. no pollution
 - C. producers internalize cost
 - D. value exceed cost
-

Q26

If monopolistic competitors earn profit, long run:

- A. consumers stop buying
- B. profits stay
- C. new firms enter, profits fall
- D. government regulates

Q27

Higher concentration ratio means:

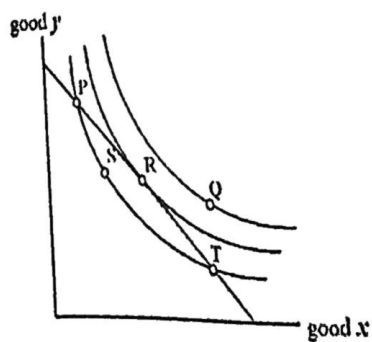
- A. more firm power
- B. more competition
- C. less competition
- D. A and C

Q28

Which is NOT correct?

- A. AFC is constant
- B. FC is constant
- C. VC changes with output
- D. ATC is U-shaped

Q29



Optimal affordable bundle is:

- A. S
- B. Q
- C. P
- D. R

Q30

Which is an implicit cost?

- A. salary to owners
- B. foregone rent on owned office
- C. cash for materials
- D. interest on loans

Part 2: Problem Solving (10 points)

Problem 1

Given the demand and supply functions for the product as follows:

$$Q_d = 140 - 20P$$

$$Q_s = 15P - 35$$

- a. Determine the market equilibrium price and quantity. Illustrate by a demand–supply diagram.
 - b. Calculate the price elasticity of demand at the market equilibrium price. State the meaning of the result.
 - c. If the government sets the product's price at \$4, will the market experience a **surplus** or a **shortage**? By how many units? Additionally, does this policy represent a **price ceiling** or a **price floor**?
 - d. The government imposes a tax of \$1 per unit on sellers. Determine the new equilibrium price and quantity.
-

Problem 2

The demand and total cost functions for King Star company are given as follows:

$$P = 45 - 0.5Q$$

$$TC = 3 + 57Q - 8Q^2 + Q^3$$

- a. Write the following cost functions: **ATC**, **VC**, **AVC**, **MC**.
- b. What price and output should the company choose to maximize its profit in the short run? Calculate the company's total profit.

- c. Find the firm's output and total profit when the government sets a ceiling price of **\$30 per unit**.
- d. What price should the company charge to maximize its revenue in the short run? Calculate the company's total revenue.